Indian Institute of Technology (IIT-Bombay)

AUTUMN Semester, 2026
COMPUTER SCIENCE AND ENGINEERING

CS230: Digital Logic Design and Computer Architecture

Tutorial - I

Full Marks: 0

Time allowed: ∞ hours

1. Simplify the following Boolean function using a Karnaugh Map and express the final result in the minimal Sum of Products (SOP) form:

$$F(A,B,C,D) = \sum m(0,2,5,7,8,10,13,15)$$

Solution: The given minterms for the 4-variable function are $m(0,2,5,7,8,10,13,15)$. Plotting these minterms on a Karnaugh map yields the following groups:

$$\text{Group 1 (Four Corners)}: m(0,2,8,10) \implies \text{simplifies to } B'D'$$
$$\text{Group 2 (Quad)}: m(5,7,13,15) \implies \text{simplifies to } BD$$

Combining the simplified terms, the minimum sum-of-products (SOP) expression is:

$$F = B'D' + BD$$

This function essentially represents an XNOR operation between variables B and D, and is completely independent of variables A and C.

$$\boxed{F = B'D' + BD}$$

2. Design a synchronous sequential circuit using D flip-flops that acts as a one-input/one-output sequence detector. The circuit should produce an output value of 1 every time the overlapping sequence **110** is detected, and 0 otherwise. Provide the state transition table and derive the minimal Sum of Products (SOP) Boolean expressions for the D flip-flop inputs and the final output.

Solution: Let the states be defined as follows to track the sequence progress:

- S_0 (State A = 00): Reset state, no valid sequence detected.
- S_1 (State B = 01): Detected the first '1'.
- S_2 (State C = 10): Detected '11'.

Using a 2-bit state assignment $y_1 y_2$ (where $S_0 = 00, S_1 = 01, S_2 = 10$), we construct the state transition and output table for input x:

Present State ($y_1 y_2$)		**Next State** ($Y_1 Y_2$)		**Output** (z)	
Name	Value	$x = 0$	$x = 1$	$x = 0$	$x = 1$
S_0 (A)	00	00	01	0	0
S_1 (B)	01	00	10	0	0
S_2 (C)	10	00	10	1	0
Unused	11	xx	xx	x	x

For D flip-flops, the excitation equations are identical to the next state equations ($D_1 = Y_1,\ D_2 = Y_2$). We extract the logic equations from the transition table using Karnaugh maps, treating the unused state 11 as don't-cares (d):

$$D_1(Y_1) = \sum m(3,5) + d(6,7) \implies D_1 = xy_2 + xy_1$$
$$D_2(Y_2) = \sum m(1) + d(6,7) \implies D_2 = xy_1'y_2'$$
$$z = \sum m(4) + d(6,7) \implies z = x'y_1$$

$$\boxed{D_1 = x(y_1 + y_2),\ D_2 = xy_1'y_2',\ z = x'y_1}$$

3. The dual of a Boolean function $F(x_1, x_2, \ldots, x_n, +, \cdot, ')$, written F^D, is the same expression as that of F with $+$ and $\cdot$ swapped. F is said to be self-dual if $F = F^D$. What are the total number of self-dual functions with n Boolean variables?

Solution:

Swapping $+ \leftrightarrow \cdot$ (and, implicitly, $0 \leftrightarrow 1$) in an expression is exactly what complementing every literal and then complementing the whole output does, by De Morgan's law. Concretely,

$$F^D(x_1, \ldots, x_n) \;=\; \overline{F(\overline{x_1}, \ldots, \overline{x_n})}.$$

So the self-duality condition $F = F^D$ becomes

$$F(x_1, \ldots, x_n) \;=\; \overline{F(\overline{x_1}, \ldots, \overline{x_n})} \qquad \text{for every input.}$$

There are 2^n possible inputs (minterms or terms in the truth table). For a self dual boolean function, if the mapping of a minterm is known, then the mapping of that input's complement can easily be determined. Therefore, the number of independent complementary pairs are -

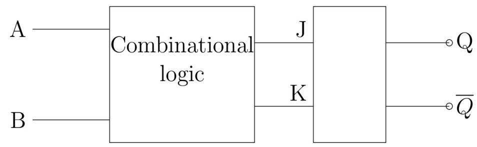
$$\frac{2^n}{2} = 2^{n-1}$$

Now, within each such pair,exactly one of the two inputs is mapped to 1 (equivalently, the other is mapped to 0).

Since there are 2^{n-1} independent pairs, each with 2 possible assignments, the total number of self-dual functions is

$$\underbrace{2 \times 2 \times \cdots \times 2}_{2^{n-1} \text{ times}} \;=\; 2^{2^{n-1}}.$$

4. Consider a circuit realization of a combinational logic block as shown in figure to obtain the following truth table:



A	**B**	$\mathbf{Q_{n+1}}$
0	0	$\overline{Q_n}$
0	1	1
1	0	Q_n
1	1	0

The combinational logic involves:

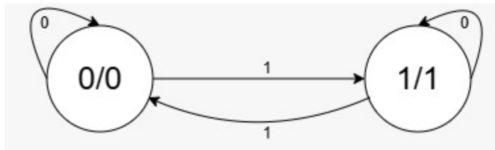
a) Only Ex-OR gates

b) AND and NOT gates

c) NOT and Ex-NOR gates

d) OR and NAND gates

5. Consider the state diagram of the Finite State Machine (FSM) provided below:



(a) Write the output sequence for the input sequence: 001010110110110111

(b) What is the functionality of the above FSM?

(c) Which flip-flop can be used to implement the above FSM without any external logical gates?

Solution:

(a) Assuming we start at state 0/0, the machine flips its state every time it sees a 1 and stays put on a 0. Since it is a Moore machine, the output is just the current state.
Input: `001010110110110111`
Output: `001100100100100101`

(b) **Functionality:** It acts as an **even parity checker**. It outputs a 1 if it has processed an odd number of 1s overall, and 0 for an even number.

(c) **Flip-flop implementation:** A **T (Toggle) flip-flop**. Its core behavior—toggling on 1 and holding on 0—matches this FSM perfectly ($Q(t+1) = T \oplus Q(t)$), meaning you can wire it up directly without extra logic gates.

6. Describe a finite state machine that will detect three consecutive coin-tosses (of one coin) that result in heads.

Solution: Let's map Heads to 1 and Tails to 0. We can build a Mealy machine that outputs a 1 when three heads in a row (111) are detected, assuming overlapping sequences are allowed.

We just need three states to track our streak:

- S_0: Reset state (streak broken).
- S_1: 1 Head detected.
- S_2: 2 consecutive Heads detected.

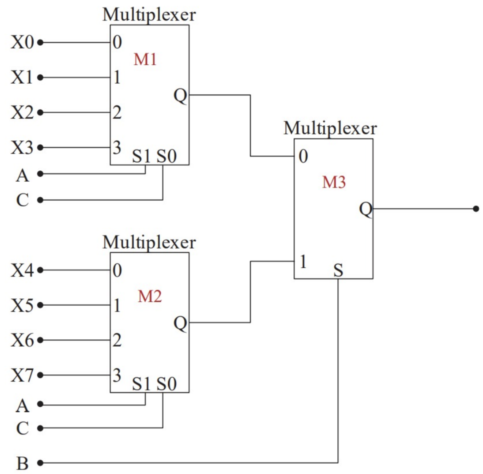
State Transition Table:

Present	**Next State**		**Output**	
State	Tail ($x = 0$)	Head ($x = 1$)	Tail ($x = 0$)	Head ($x = 1$)
S_0	S_0	S_1	0	0
S_1	S_0	S_2	0	0
S_2	S_0	S_2	0	1

Notice that if we are in S_2 and get another Head, we output a 1 (success!) and loop right back to S_2 to catch overlapping sequences (e.g., getting 4 heads in a row outputs 1 twice). Any Tail immediately ruins the streak, sending us straight back to S_0.

7. What values of $(X_0, X_1, X_2, X_3, X_4, X_5, X_6, X_7)$ will realize the Boolean function:

$$\overline{A} + \overline{A} \cdot \overline{C} + A \cdot \overline{B} \cdot C \ ?$$



Solution

(a) Circuit Analysis

The circuit in figure above implements an 8×1 multiplexer using two 4×1 multiplexers and one 2×1 multiplexer. Based on the connections:

- The 2×1 MUX (M3) is controlled by B, making B the Most Significant Bit (MSB), S_2.
- The 4×1 MUXes (M1 and M2) are controlled by A and C. Since A is connected to S_1 and C is connected to S_0, C is the Least Significant Bit (LSB).
- The final select line order is $S_2 S_1 S_0 = BAC$.

(b) Simplify the Boolean Expression The given Boolean function is:

$$F = \overline{A} + \overline{A} \cdot \overline{C} + A \cdot \overline{B} \cdot C$$

By applying the absorption law ($\overline{A} + \overline{A} \cdot \overline{C} = \overline{A}$), we get:

$$F = \overline{A} + A \cdot \overline{B} \cdot C$$

By applying the distributive law/consensus theorem ($X + \overline{X} \cdot Y = X + Y$), the expression simplifies to:

$$F = \overline{A} + \overline{B} \cdot C$$

(c) Find the Minterms Since our select lines are in the order BAC, we need to expand the simplified expression into standard Sum of Products (SOP) form based on the variables B, A, and C:

$$\begin{aligned}\overline{A} &= \overline{A} \cdot (B + \overline{B}) \cdot (C + \overline{C}) \\ &= \overline{B} \cdot \overline{A} \cdot \overline{C} + \overline{B} \cdot \overline{A} \cdot C + B \cdot \overline{A} \cdot \overline{C} + B \cdot \overline{A} \cdot C \\ &= m_0 + m_1 + m_4 + m_5\end{aligned}$$

$$
\begin{aligned}
\overline{B} \cdot C &= \overline{B} \cdot C \cdot (A + \overline{A}) \\
&= \overline{B} \cdot \overline{A} \cdot C + \overline{B} \cdot A \cdot C \\
&= m_1 + m_3
\end{aligned}
$$

Combining the minterms (and omitting duplicates), we get:

$$F(B, A, C) = \Sigma m(0, 1, 3, 4, 5)$$

(d) Final Answer For an 8×1 MUX, the minterms directly correspond to the data inputs X_i that must be connected to logic 1. Therefore, the inputs X_0, X_1, X_3, X_4, and X_5 are 1, while X_2, X_6, and X_7 are 0.

$$(X_0, X_1, X_2, X_3, X_4, X_5, X_6, X_7) = (1, 1, 0, 1, 1, 1, 0, 0)$$

8. Design a combinational circuit to realize the following function, where $X = x_2x_1x_0$ is a 3-bit binary input and $F(X) = F_2F_1F_0$ is the 3-bit binary output:

$$\boxed{F(X) = \begin{cases} X + 1, & 0 \le X \le 3, \\ X - 1, & 4 \le X \le 7. \end{cases}}$$

Determine the minimal SOP and POS.

Solution

(a) Truth Table

x_2	x_1	x_0	X	$F(X)$	F_2	F_1	F_0
0	0	0	0	1	0	0	1
0	0	1	1	2	0	1	0
0	1	0	2	3	0	1	1
0	1	1	3	4	1	0	0
1	0	0	4	3	0	1	1
1	0	1	5	4	1	0	0
1	1	0	6	5	1	0	1
1	1	1	7	6	1	1	0

(b) Minimal SOP – K-map Minimization

$$F_2 = \Sigma m(3, 5, 6, 7)$$

	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$(3, 7) \to x_1x_0, \qquad (5, 7) \to x_2x_0, \qquad (6, 7) \to x_2x_1$$

$$\boxed{F_2 = x_1x_0 + x_2x_0 + x_2x_1}$$

$F_1 = \Sigma m(1, 2, 4, 7)$

	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$1 \to x_2' x_1' x_0, \qquad 2 \to x_2' x_1 x_0', \qquad 4 \to x_2 x_1' x_0', \qquad 7 \to x_2 x_1 x_0$$

$$\boxed{F_1 = x_2' x_1' x_0 + x_2' x_1 x_0' + x_2 x_1' x_0' + x_2 x_1 x_0}$$

$F_0 = \Sigma m(0, 2, 4, 6)$

	00	01	11	10
0	1	0	0	1
1	1	0	0	1

The wrap-around group $(0, 2, 4, 6)$ gives x_0'.

$$\boxed{F_0 = x_0'}$$

Therefore,

$$\boxed{F_2 = x_1 x_0 + x_2 x_0 + x_2 x_1}, \qquad \boxed{F_1 = x_2' x_1' x_0 + x_2' x_1 x_0' + x_2 x_1' x_0' + x_2 x_1 x_0}, \qquad \boxed{F_0 = x_0'}.$$

(c) Minimal POS – K-map Minimization
For POS minimization, group the 0's.

$F_2 = \Pi M(0, 1, 2, 4)$

	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$(0, 1) \to (x_2 + x_1), \qquad (0, 2) \to (x_2 + x_0), \qquad (0, 4) \to (x_1 + x_0)$$

$$\boxed{F_2 = (x_2 + x_1)(x_2 + x_0)(x_1 + x_0)}$$

$F_1 = \Pi M(0, 3, 5, 6)$

	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$0 \to (x_2 + x_1 + x_0), \qquad 3 \to (x_2 + x_1' + x_0'), \qquad 5 \to (x_2' + x_1 + x_0'), \qquad 6 \to (x_2' + x_1' + x_0)$$

$$\boxed{F_1 = (x_2 + x_1 + x_0)(x_2 + x_1' + x_0')(x_2' + x_1 + x_0')(x_2' + x_1' + x_0)}$$

$F_0 = \Pi M(1, 3, 5, 7)$

	00	01	11	10
0	1	0	0	1
1	1	0	0	1

The group $(1, 3, 5, 7)$ gives x_0'.

$$\boxed{F_0 = x_0'}$$

Therefore,

$$\boxed{F_2 = (x_2 + x_1)(x_2 + x_0)(x_1 + x_0)}$$

$$\boxed{F_1 = (x_2 + x_1 + x_0)(x_2 + x_1' + x_0')(x_2' + x_1 + x_0')(x_2' + x_1' + x_0)}$$

9. Design a synchronous sequential circuit using two flip-flops Q_1 and Q_0, and one control input x. The state is represented as Q_1Q_0, where Q_1 is the **MSB** (leftmost bit) and Q_0 is the **LSB** (rightmost bit).

For $x = 0$, the required sequence is

$$\boxed{00 \to 01 \to 10 \to 11 \to 00 \to \cdots}$$

and for $x = 1$,

$$\boxed{00 \to 11 \to 10 \to 01 \to 00 \to \cdots}$$

Determine the required flip-flop inputs for:

(a) D flip-flop realization: D_1, D_0

(b) T flip-flop realization: T_1, T_0

Solution

(a) State Transition Table

Present State		$x=0$		$x=1$	
Q_1	Q_0	Q_1^+	Q_0^+	Q_1^+	Q_0^+
0	0	0	1	1	1
0	1	1	0	0	0
1	0	1	1	0	1
1	1	0	0	1	0

Equivalently,

Q_1	Q_0	x	Q_1^+	Q_0^+
0	0	0	0	1
0	0	1	1	1
0	1	0	1	0
0	1	1	0	0
1	0	0	1	1
1	0	1	0	1
1	1	0	0	0
1	1	1	1	0

(b) D Flip-Flop Realization
For a D flip-flop,

$$\boxed{D = Q^+}$$

The D flip-flop excitation table is

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

Hence,

$$D_1 = Q_1^+, \qquad D_0 = Q_0^+.$$

For D_1,

$$D_1 = \Sigma m(1,2,4,7)$$

and therefore

$$\boxed{D_1 = Q_1 \oplus Q_0 \oplus x}$$

or, in SOP form,

$$\boxed{D_1 = Q_1'Q_0'x + Q_1'Q_0x' + Q_1Q_0'x' + Q_1Q_0x}$$

For D_0,

$$D_0 = \Sigma m(0,1,4,5)$$

and therefore

$$\boxed{D_0 = Q_0'}$$

Thus,

$$\boxed{D_1 = Q_1 \oplus Q_0 \oplus x, \qquad D_0 = Q_0'}$$

(c) T Flip-Flop Realization
For a T flip-flop,

$$\boxed{Q^+ = T \oplus Q} \qquad \Longrightarrow \qquad \boxed{T = Q \oplus Q^+}$$

The T flip-flop excitation table is

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

Using $T_1 = Q_1 \oplus Q_1^+$ and $T_0 = Q_0 \oplus Q_0^+$, the required inputs are obtained as follows:

Q_1	Q_0	x	Q_1^+	Q_0^+	T_1	T_0
0	0	0	0	1	0	1
0	0	1	1	1	1	1
0	1	0	1	0	1	1
0	1	1	0	0	0	1
1	0	0	1	1	0	1
1	0	1	0	1	1	1
1	1	0	0	0	1	1
1	1	1	1	0	0	1

Thus,

$$T_1 = \Sigma m(1,2,5,6)$$

and

$$\boxed{T_1 = Q_0 \oplus x}$$

Similarly,

$$T_0 = \Sigma m(0, 1, 2, 3, 4, 5, 6, 7)$$

and therefore

$$\boxed{T_0 = 1}$$

Hence, the required T flip-flop inputs are

$$\boxed{T_1 = Q_0 \oplus x, \qquad T_0 = 1}$$

(d) Final Answer

Flip-Flop Type	MSB Flip-Flop Q_1	LSB Flip-Flop Q_0
D	$D_1 = Q_1 \oplus Q_0 \oplus x$	$D_0 = Q_0'$
T	$T_1 = Q_0 \oplus x$	$T_0 = 1$

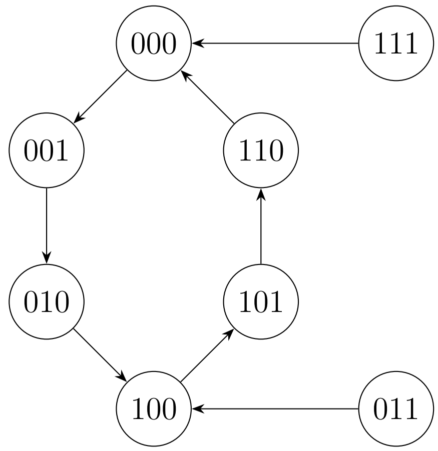
with

$$\boxed{Q_1 Q_0 = \text{MSB LSB}}$$

and Q_0 being the rightmost bit.

10. Design a synchronous counter that follows the state transitions specified by the given state diagram, using D flip-flops.

Given State Diagram



Solution

(a) State Transition Table

Present State			Next State		
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	1	0	0	0

For a D flip-flop,

$$\boxed{D_i = Q_i^+}$$

Therefore,

$$D_2 = Q_2^+, \qquad D_1 = Q_1^+, \qquad D_0 = Q_0^+.$$

(b) D Flip-Flop Input Equations
From the transition table,

$$D_2 = \Sigma m(2, 3, 4, 5)$$

$$D_1 = \Sigma m(1, 5)$$

$$D_0 = \Sigma m(0, 4)$$

After Boolean minimization,

$$\boxed{D_2 = Q_1 \oplus Q_2}$$

$$\boxed{D_1 = Q_0 Q_1'}$$

$$\boxed{D_0 = Q_0' Q_1'}$$

Hence,

$$\boxed{D_2 = Q_1 \oplus Q_2, \qquad D_1 = Q_0 Q_1', \qquad D_0 = Q_0' Q_1'}$$

11. Realize the following Boolean function using one $4:1$ multiplexer and at most one additional gate.

$$\boxed{F(A,B,C) = \Sigma m(1,2,5,6)}$$

Use A and B as the select inputs of the multiplexer.

Determine the required MUX data inputs I_0, I_1, I_2, I_3 and draw the resulting circuit.

Solution

For a $4:1$ MUX, choose A and B as the select inputs:

$$S_1 = A, \qquad S_0 = B.$$

Using Shannon expansion with respect to A and B,

$$F = \bar{A}\bar{B}\,F_{00} + \bar{A}B\,F_{01} + A\bar{B}\,F_{10} + AB\,F_{11},$$

where

$$F_{ij} = F(A=i, B=j, C).$$

Given

$$F(A,B,C) = \Sigma m(1,2,5,6),$$

we obtain each cofactor by substituting the corresponding values of A and B:

$$\begin{aligned} F_{00} &= F(0,0,C) = \Sigma m(1) = C, \\ F_{01} &= F(0,1,C) = \Sigma m(2) = \bar{C}, \\ F_{10} &= F(1,0,C) = \Sigma m(5) = C, \\ F_{11} &= F(1,1,C) = \Sigma m(6) = \bar{C}. \end{aligned}$$

Therefore,

$$F = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}.$$

Comparing this with the $4:1$ MUX equation,

$$F = \bar{A}\bar{B}I_0 + \bar{A}BI_1 + A\bar{B}I_2 + ABI_3,$$

we obtain

$$\boxed{I_0 = C, \qquad I_1 = \bar{C}, \qquad I_2 = C, \qquad I_3 = \bar{C}}$$

Thus,

$$\boxed{S_1 = A, \quad S_0 = B, \quad I_0 = C,\ I_1 = \bar{C},\ I_2 = C,\ I_3 = \bar{C}}$$

where $\bar{C}$ is generated using the one additional NOT gate.
